

A Replication of Study 3 in Stellar et al. (2018): Awe and Humility

Tizian M. B. Schuhbeck

Department of Psychology, Ludwig-Maximilians-Universität München

”Forschungsorientiertes Praktikum I – Grundlagen der Forschung”

Prof. Dr. Felix Schönbrodt

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Author Note

Tizian M. B. Schuhbeck  <https://orcid.org/0009-0005-5903-674X>

Correspondence concerning this article should be addressed to Tizian M. B. Schuhbeck,

Email: t.schuhbeck@campus.lmu.de

Abstract

As part of the CREP project... we follow a direct replication and extension approach.

Impact Statement

This template helps you get started.

Keywords: Registered Replication, Humility, Awe, Emotion, Self-Concept, CREP, Video Manipulation

A Replication of Study 3 in Stellar et al. (2018): Awe and Humility

The seminal findings of the Open Science Collaboration (2015) triggered what is now known as the ‘replication crisis’ in psychological science, prompting researchers to question thusfar widely unchallenged publications by and large. Since then, there has been an ongoing effort within the psychological research community to actively persue replication of affected studies with important findings, aiming to reevaluate their validity and ultimately restore trust in the body of psychological knowledge. As part of the Collaborative Replications and Education Project (CREP) project of Grahe et al. (2026), we aim to contribute to this endeavour by investigating the replicability of the results in ‘Study 3’ of Stellar et al. (2018): Awe and Humility. In this study, the authors hypothesize momentary experiences of awe to cause a temporary increase in humility. The authors operationalize this hypothesis by means of a randomized video intervention, designed to induce either a state of awe or a neutral state, with a subsequent evaluation on a behavioral measure of humility. Both theoretical background and original study design are outlined in detail in the original publication of Stellar et al. (2018). To ensure transparency, following the general recommendation of adopting the principles of open science (Schönbrodt et al., 2018), we have made all relevant materials for this replication study available on the platform Open Science Framework (OSF), in particular the preregistration form at <https://osf.io/537aq>.

A Direct Replication and Extension of Stellar et al. (2018)

In this replication study, we aim to perform a direct replication of study 3 Stellar et al. (2018), adopting the authors orignial methodology and analysis to the best of our knowledge and possibilities, while further – in addition – extending the study design to allow for analyses of supplementary hypotheses, taking care to ensure the orginial study remains intact as an isolated part of our extended study design. As our recruitment of participants has taken place in germany, however, we have opted to translate the study material into german, in particular the audio track for the neutral condition video. Our recruitment process follows the snowball sampling principle (Goodman, 1961), so consequently our sample consits of mostly students, as well as their friends and relatives, resulting in demographic range that differs from to the original sample of Stellar et

al. (2018). All known deviations are documented in detail within the preregistration form. For the direct replication, we aim to replicate the following hypothesis.

H1: Relative to a neutral control condition, the original awe inducing video increases humility in a behavioral measure.

We extend the replication study by introducing a third condition, in which participants are shown a more modern video, which is also designed to induce awe. Based on this condition, we additionally investigate the following confirmatory hypothesis.

H2: Relative to a neutral control condition, the new awe inducing video increases humility in a behavioral measure.

Furthermore, we aim to address the natural follow-up question of how the two experimental conditions compare by means of the following exploratory hypothesis.

E1: The new (more modern) awe inducing video leads to higher awe ratings compared to the old awe inducing video.

Note, that as documented in the preregistration, we have formulated a total of six supplementary exploratory hypotheses. Of these, E1 was chosen at random prior to any further data analyses. The full list of all exploratory hypotheses is documented in the publicly available preregistration form.

Method

The scripts required for reproducing the following results have been made publically available on GitHub at https://github.com/TiziSchuh/Stellar_Rep_01. The hypotheses, operationalizations, analysis plan, inference criteria, and sample size justification have been preregistered at <https://osf.io/537aq>. Open material and data can all be found at <https://osf.io/cy287>.

Participants

We note that in the following, we described, analysed and drew conclusions from interim data, which was collected from 19.06.26 to 10.07.26 and consisted of 189 participants before exclusions. We preregistered to exclude participants who did not write anything in the writing assignment (26 exclusions) or indicated that their data should not be used (1 exclusions), leading

to a final sample size of $n = 162$, resulting in conditions being distributed 50 (neutral video), 58 (original awe video) and 54 (modern awe video). The final sample had a mean age of 32 ($SD = 16.2$, range: 18–77). 112 participants reported identifying as female, 46 as male and 3 as non-binary, with 1 participant opting not to report their gender identity.

Measures and Materials

We operationalized our hypotheses by measuring three variables. For the Independent Variable (IV), we measured the randomized video condition with levels “neutral”, “awe original” and “awe modern”. We then measured awe as an emotional state on a behavioral scale as the mean of the three items “Staunen”, “Faszination”, with responses on a seven point likert scale each.

For the Dependent Variable (DV), we measured the state of humility as the log-transformed ratio of listed personal strengths to weaknesses. The number of strengths and weaknesses are counted by two independent raters from the free-text answer of the participants in a writing exercise. We use the “Codebook Instructions for Counting Participant-Reported Strengths and Weaknesses” from <https://doi.org/10.26443/msurj.v19i1.231> (see Appendix B).

Link to code-book!!!

Procedure

Our study is a one-time, cross-sectional online study. Participants are randomly assigned one of the three video conditions. We have provided an overview of the general steps within the study procedure in Figure 1. First, based on their respective condition, they will watch a corresponding video. Subsequently, they will be asked to do a writing exercise, describing their strengths and weaknesses. After the exercise, they report on their emotional state. Finally they report on the control questions. The entire study material – in particular the respective FormR files – can be found on OSF.

Statistical Procedure

For H1, we mirrored the statistical procedure applied by Stellar et al. (2018). To ensure approximate normal-distribution of the DV (humility), we transformed the data by using the

natural logarithm¹, i.e. the final humility score for each participant i was calculated as

$$y_{\text{humility},i} = \ln \left(\frac{N_{\text{strengths},i}}{N_{\text{weaknesses},i}} \right).$$

This necessitated the exclusion of participants who did not write down anything that was counted as a weakness according to the code-book. Together with the removal of outliers, characterized by the number of strengths or weaknesses being more than 3 SDs over the mean, the final sample for H1 had $n=73$ valid participants. Using this sample, we calculated a one-sided two-group t-test comparing the log-transformed balance score of the neutral video vs the awe original video at significance level $\alpha=.05$. For H2, we applied the same transformation and exclusions described above, yielding $n=76$ valid participants for the final sample, for which we calculated a two-group t-test comparing the log-transformed balance score of the neutral video vs the awe new video.

For E1, we calculated the Cohen's d effect size (Cohen, 1988) comparing the awe-score averages in the original video condition to the new video condition. maybe significance test...

Deviations from Preregistration

Since the final sample for H1 fell short of the 100 participants aimed for, we only achieved a power of .696, rather than the desired power of .80 or greater.

Results

On the IV (awe), we observed the mean=4.12 and SD=1.85 after pre-analysis exclusions. On the DV (humility), we observed the mean=0.59 with SD=0.48 for the sample of H1, and the mean=0.66 with SD=0.48 for the sample of H2.

Confirmatory Results

For H1, the result of the t-test was $\Delta M = 0.09$, 95% CI $[-0.10, \infty]$, $t(71) = 0.80$, $p = .212$. Based on this, we reject the alternative hypothesis (the original awe video increases humility relative to the neutral condition) given the p-value $p=0.212$ exceeding the significance

¹ We preregistered the use of the “natural” logarithm, while at the same time – erroneously – denoting the R-function “log2” in the final line of the paragraph. In the actual analysis, we have used the correct R-function “log” instead.

level $\alpha=.05$.

For H2, the result of the t-test was $\Delta M = -0.03$, 95% CI $[-0.22, \infty]$, $t(74) = -0.26$, $p = .602$. Based on this, we reject the alternative hypothesis (the new awe video increases humility relative to the neutral condition) given the p-value $p=0.602$ exceeding the significance level $\alpha=.05$.

Exploratory Results

Calculating Cohen's d for E1 (comparing new and old video w.r.t. inducing high levels of awe) yielded an effect size of -0.03 with 95% CI $[-0.4, 0.34]$.

Discussion

In this paper, we provided a direct replication and extension of 'Study 3' in Stellar et al. (2018). We investigated the original hypothesis of momentary experiences of awe inducing humility on a behavioral measure. Furthermore, we introduced an alternative modern video condition, comparing it to the original video condition. We were able to replicate and extend the study, however, we were not able to recreate the results in the original study – our data did not provide evidence for the alternative hypothesis in H1. The new video condition (as part of our extension) neither resulted in the desired effect – our data also did not provide evidence for the alternative hypothesis in H2. Furthermore, based on the effect calculated for E1, there seem to be no substantial differences in the effectiveness of inducing awe between the original and old video beyond sampling variability. All in all, our results do not support the findings of Stellar et al. (2018).

Limitations and Future Directions

Although we aimed for a direct replication to the best of our knowledge and possibilities, there are some discrepancies to keep in mind. Most notably, the participants for our replication were recruited using a snowball sampling approach. As a result, the age in our sample was substantially lower on average. Especially in conjunction with the fact, that our replication has taken place more than eight years after the publication of Stellar et al. (2018), it may be possible, that participants in our sample are generally more used to watching such videos, possibly resulting in a substantial reduction of the actual amount of awe being induced, compared to participants in

the original study. The similarly high levels on the self-reported awe scale may have simply been positively skewed by acquiescence. In the future, alternative ways to induce awe may be explored. Furthermore, as we have translated the awe scale into German, we cannot a priori rule out possible measurement non-invariances, which could also causally explain why the effect was not reproduced. This may be subject to further investigation in future replication studies.

Finally, the fact that we only achieved a power of around .70 arguably forms the biggest limitation of our replication. A well known fact in empirical research is that “lack of evidence is not evidence of absence” (Altman & Bland, 1995). Hence, based solely on the finding presented here, we cannot and should not rule out, that the power was simply not big enough to detect the investigated effect. This limitation, however, already being addressed precisely by the ongoing CREP project, as well as previous replications of this study (Bertrand et al., 2024).

Conclusion

Essenz in 2-3 Sätzen.

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Figure 1

An Illustration of the Study Procedure.

